Lab Tasks

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Course: Artificial Intelligence Lab (CS - 325L)

Program: (BS) - Computer Science

Lab #14

Open Ended Lab could be based on implementation of PYTHON or PROLOG programming.

Implement any scenario of your choice. Complete a task which is different from those taught in class/laboratory or a similar model with enhancements/improvements clearly demonstrated. You can select any specific technique or a model from a research paper. The details should be well documented. Codes should be with comments.

PROLOG
Activity- 12:
Create a basic calculator prolog program for making basic calculations (add, subtract, multiply, divide, average). Use separate goal statements for making each calculation (with constant numeric values).

```
% Activity 12
% Addition
add :-
   write('Enter first number: '), read(A),
   write('Enter second number: '), read(B),
   Result is A + B,
   write('Addition: '), write(Result), nl.

% Subtraction
subtract :-
   write('Enter first number: '), read(A),
   write('Enter second number: '), read(B),
   Result is A - B,
   write('Subtraction: '), write(Result), nl.

% Multiplication
multiply :-
   write('Enter first number: '), read(A),
   write('Enter second number: '), read(B),
   Result is A * B,
   write('Multiplication: '), write(Result), nl.

% Division
divide :-
   write('Enter numerator: '), read(A),
   write('Enter denominator: '), read(B),
   (   B =\= 0
   ->  Result is A / B,
       write('Division: '), write(Result), nl
   ;   write('Error: Division by zero is not allowed.'), nl
   ).

% Average of three numbers
average :-
   write('Enter first number: '), read(A),
   write('Enter second number: '), read(B),
   write('Enter third number: '), read(C),
   Avg is (A + B + C) / 3,
   write('Average: '), write(Avg), nl.
```

```
| ?- add.
Enter first number: 2.
Enter second number: 3.
Addition: 5

(16 ms) yes
| ?- subtract.
Enter first number: 3.
Enter second number: 1.
Subtraction: 2

yes
| ?- multiply.
Enter first number: 3.
Enter second number: 3.
Multiplication: 9

(15 ms) yes
| ?- divide.
Enter numerator: 2.
Enter denominator: 6.
Division: 0.33333333333333331

(15 ms) yes
| ?- average.
Enter first number: 2.
Enter second number: 3.
Enter third number: 4.
Average: 3.0

yes
```

Activity- 13:

Make a calculator program by using write() and read() predicates, for taking numeric input from user and displaying text after program execution.

```
% Activity 13
calculator :-
    write('Enter the first number: '), read(A),
    write('Enter the second number: '), read(B),

    Sum is A + B,
    Diff is A - B,
    Prod is A * B,

    (   B =\= 0
    ->  Div is A / B
    ;   Div = 'undefined (division by zero)'
    ),

    Avg is (A + B) / 2,

    nl, write('--- Calculation Results ---'), nl,
    write('Addition: '), write(Sum), nl,
    write('Subtraction: '), write(Diff), nl,
    write('Multiplication: '), write(Prod), nl,
    write('Division: '), write(Div), nl,
    write('Average: '), write(Avg), nl.
```

```
| ?- calculator.
Enter the first number: 2.
Enter the second number: 1.

--- Calculation Results ---
Addition: 3
Subtraction: 1
Multiplication: 2
Division: 2.0
Average: 1.5

yes
```

PYTHON

Activity- 4:

If you could invite anyone to dinner. Make a list that includes at least three people you'd like to invite to dinner. Then use your list to print a message to each person, inviting them to dinner.

```python
print("Activity 4:")
guests = ['Hamza', 'Sara', 'Ahmed']
for guest in guests:
    print(f"Dear {guest}, you're invited to dinner at my place!")
print()
```

```
Activity 4:
Dear Hamza, you're invited to dinner at my place!
Dear Sara, you're invited to dinner at my place!
Dear Ahmed, you're invited to dinner at my place!
```

Activity- 5:

Changing Guest List: You just heard that one of your guests can't make the dinner, so you need to send out a new set of invitations. You'll have to think of someone else to invite.

- **Modify your list, replacing the name of the guest who can't make it with the name of the new person you are inviting.**
- **Print a second set of invitation messages, one for each person who is still in your list.**

```python
print(f"{guests[1]} can't make it to dinner.")
guests[1] = 'Bilal'
for guest in guests:
    print(f"Dear {guest}, you're still invited to dinner at my place!")
print()
```

```
Activity 5:
Fatima can't make it to dinner.
Dear Hamza, you're still invited to dinner at my place!
Dear Bilal, you're still invited to dinner at my place!
Dear Ahmed, you're still invited to dinner at my place!
```

Activity- 6:

Write a program to read 6 numbers and create a dictionary having keys EVEN and ODD.
Dictionary's value should be stored in list. Your dictionary should be like: {'EVEN':[8,10,64], 'ODD':[1,5,9]}

```
print("Activity 6:")
numbers = []
print("Enter 6 numbers:")
for _ in range(6):
    numbers.append(int(input()))
result = {'EVEN': [], 'ODD': []}
for n in numbers:
    if n % 2 == 0:
        result['EVEN'].append(n)
    else:
        result['ODD'].append(n)
print(result)
print()
```

```
Activity 6:
Enter 6 numbers:
 1
 2
 6
 7
 3
 4
{'EVEN': [2, 6, 4], 'ODD': [1, 7, 3]}
```

Activity- 7:

Write a definition of a method count_now(places) to find and display those place names, in which there are more than 5 characters.

```
def count_now(places):
    print("Activity 7:")
    for place in places:
        if len(place) > 5:
            print(place)

count_now(['London', 'Tokyo', 'Istanbul', 'Karachi', 'New York'])
print()
```

```
Activity 7:
London
Istanbul
Karachi
New York
```